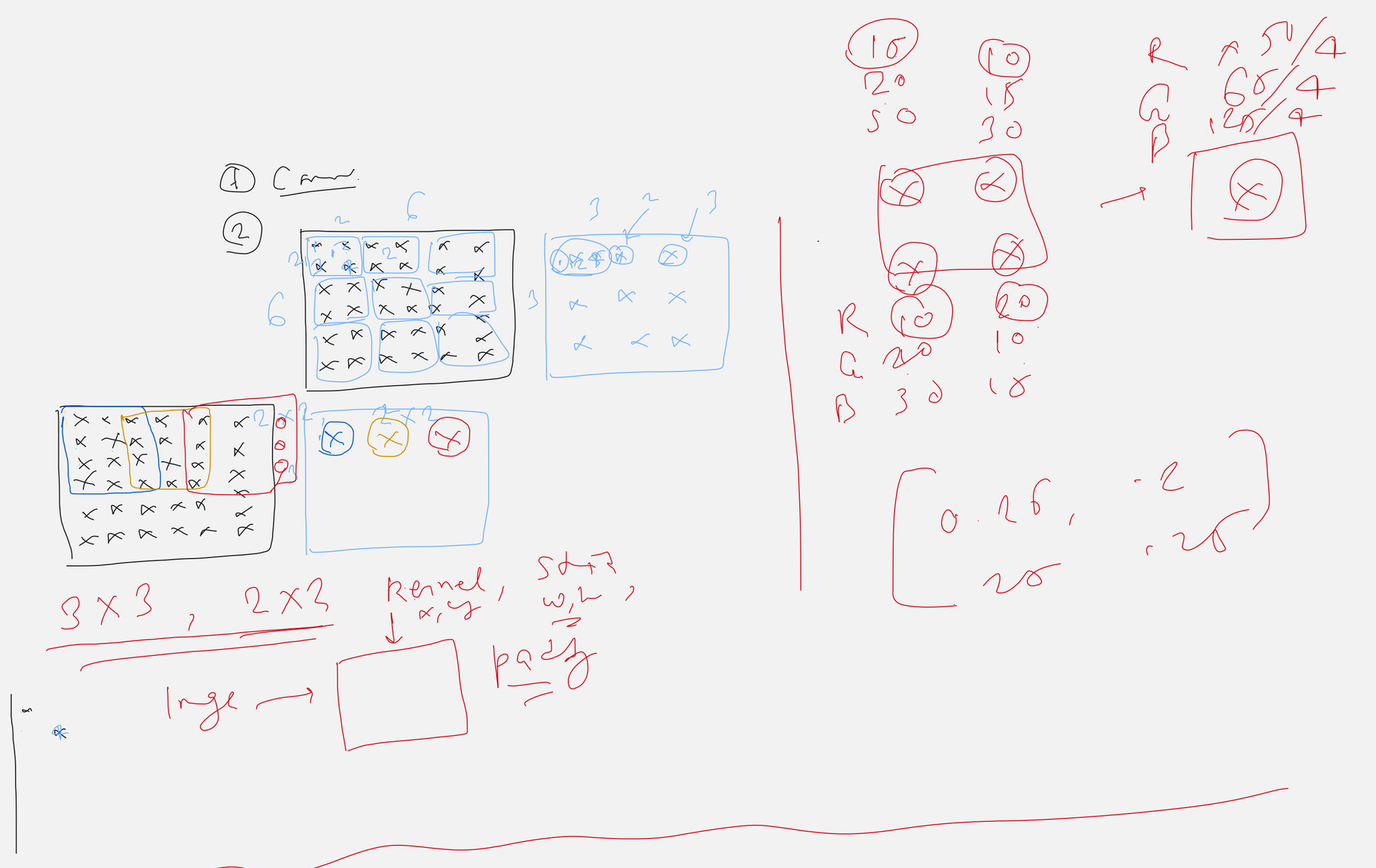
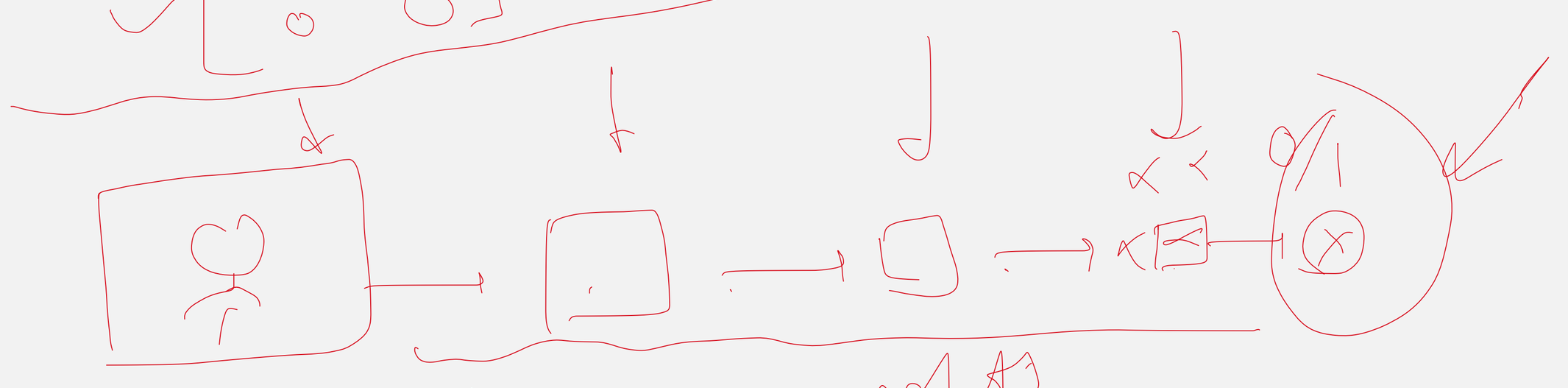

① Cnn
②
3×3 , 2×2 kernel, stride, padding
Image
R 50/4
G 65/4
B 125/4
R G B
0.25, .2
.25, .25

Generat~ to weighted Sum

$$\begin{bmatrix} .25 & .2 \\ .4 & .8 \end{bmatrix} \to$$

$$✓\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \to$$

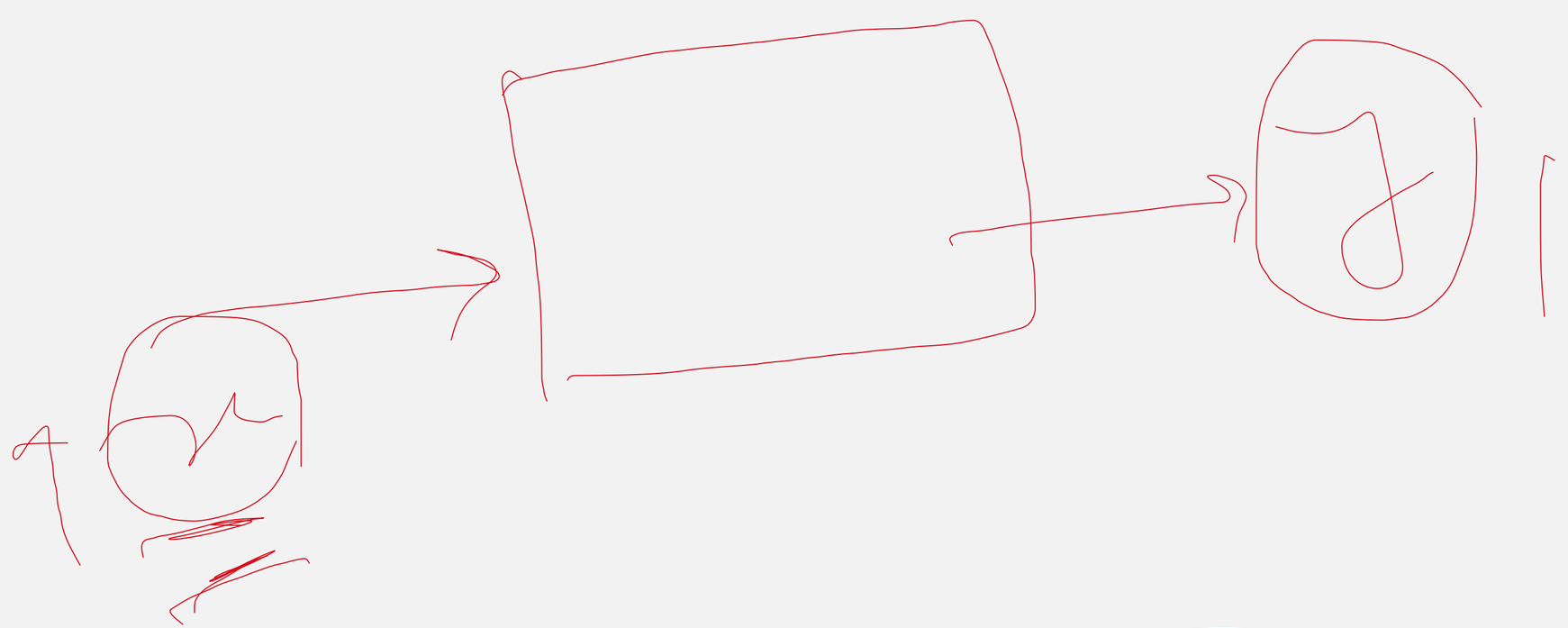


Find these weights for which M/F prediction…

$$\frac{dy}{dx} = ?$$

– Slope

– rate of change of y wrt x



$$\frac{dy}{dx} = \text{ROI}$$

$x = 100, \quad y = 200$

$x = 100.01, \quad y = 201$

$$\frac{dy}{dx} = \frac{201 - 200}{100.01 - 100} = \boxed{100}$$

$$x \to \boxed{f} \to y, \quad y = f(x)$$

$\frac{dy}{dx}$, at $x = 100$

eps = 0.000001

$y_1 = f(\cancel{Dx}) = f(100)$

$y_2 = f(100 + eps)$

$$\frac{dy}{dx} = \frac{y_2 - y_1}{eps}$$

$y = 4x^2 - 3x + 10$

Find Minima.

$\boxed{\frac{dy}{dx}} = 8x - 3, \quad x = \frac{3}{8}$

$\frac{d^2y}{dx^2} = \boxed{8}$ Minima

y

$y = 10x - x^2 - 2$

$\frac{dy}{dx} = 10 - 2x = 0$

$x = 5$

$\frac{d^2y}{dx^2} = \boxed{-2}$ | Maxima